13. Intangible assets continued

Goodwill has been allocated to three CGUs, which align with the reportable operating segments, as follows:

	2019 $ million	2018 $ million
Networks & Security	73.4	72.0
Lifecycle Service Assurance	37.6	37.6
Connected Devices	46.1	46.1
	157.1	155.7

Annual impairment test

The Group has an annual impairment testing date of 30 November. The key assumptions used in the value in use calculations were:

- revenue growth rates;
- gross margin;
- operating expenses;
- discount rate; and
- growth rate used to extrapolate cash flows beyond the five-year period covered by management's projections.

The cash flows are derived from the most recent financial budgets for the next financial year, as approved by management, and the Group's three-year strategic plan. Cash flows in years four and five are extrapolated based on long range plans. Cash flows in subsequent years have been extrapolated using a steady 2.5 per cent for all CGUs (2018 2.5 per cent for all CGUs), which management estimates to be the approximate average long-term growth rate for the industries in which these units operate. Fundamentally, this long-term growth is based on a proxy for global long-term inflation taking into consideration more developed and developing markets. The growth rates used in the value in use calculations are set at the same level for each CGU as all the CGUs operate within similar markets which share the same growth drivers and characteristics. The discount rates incorporate the specific risks relating to each CGU.

The discount rate applied to the cash flows is based on the weighted average cost of capital of comparable companies by taking the risk free rate for 30-year government bonds and making an adjustment to reflect the increased risk of investing in equities. In making this adjustment, the inputs required are the equity market risk premium, beta, and the risk adjustment applied to reflect the systematic risk of Spirent and the specific CGUs, taking into account factors such as size and the territories in which each CGU operates.

The cash flows have been discounted using the following pre-tax discount rates:

	2019 %	2018 %
Networks & Security	13.8	15.1
Lifecycle Service Assurance	13.5	14.3
Connected Devices	13.1	14.8

For Spirent the key factor in relation to the cash flow forecasts is the ability to forecast revenue. All CGUs operate in the data technology market and generate a high gross profit (gross margin); consequently changes in revenue can have a significant impact on the operating profit and cash flows. Revenue growth rates used in the projections are based on management's estimate of growth in the markets served and take into account historical levels of growth, expected future developments in products and technology, industry forecasts and macro-economic conditions in the territories in which the CGUs operate. Gross margin and operating expenses are based on historical values adjusted for the effect of revenue growth and cost reduction actions committed prior to the impairment testing date.

At Networks & Security, Cloud and IP is expected to maintain its leadership position in high-speed Ethernet and this, together with optimised 400G volumes and growth in network virtualisation, is expected to drive earnings. Further growth in Networks & Security is expected in the Positioning business, with focus on core markets in APAC and on the emerging autonomous vehicle market. Management expects that the security business will benefit in the longer term from the launch of data breach emulation, and also the move to a subscription model and further expansion of SecurityLabs into EMEA and APAC. Cyber Security is expected to benefit from synergies with Positioning and continued expansion in complementary solutions with Cloud and IP.

The continuing drive in lab sales and enhancements to the feature set at the Mobility Infrastructure business unit is expected to grow revenue in Lifecycle Service Assurance on a relatively flat cost base in the near term. Management expects revenue increases at the Customer Experience Management business unit driven by the VisionWorks sales strategy and a relatively flat gross margin over the three-year forecast period. The Lumos legacy business in Lifecycle Service Assurance is continuing to drop in the near term and is being replaced by growth in new Ethernet products supported by investment in product development, particularly in virtualisation, mobility testing and 5G. The individual business units in Lifecycle Service Assurance are expected to work together to deliver the three-year plan to meet changes in customer needs and the competitive landscape. A new General Manager has recently been appointed at Lifecycle Service Assurance and there is increased emphasis on Testing-as-a-Service and development of the services and solutions businesses.